

GESTURE RECOGNITION Beckons The Next Wave Of UI



Dilin Anand is a senior assistant editor at EFY. He is B.Tech from University of Calicut, currently pursuing MBA from Christ University, Bengaluru

User interface (UI) design problem: Too many steps to do something. Engineers have been enabling advances here by moving from physical keys, switches and knobs to resistive touch-screens and later to capacitive screens that accept multi-touch inputs, swipes and other gestures. Also seen are various wearable controllers and vision based systems that let you control electronics without laying a finger on a physical device.

It is now six years since Microsoft disrupted gaming with optical sensors in its Kinect, and gesture technology looks ready to take a leap in 2016.

Swiping your way through screens

It all starts with recognising touch, swipes and gestures. GestureTek is a company that has been into 3D video gesture control since the early 2000s. They have been keeping up with display technology, too, and now offer 4K displays with multi-touch enabled on these. Their multi-touch tables offer up to 32 touch points on 213cm displays.

They also have GestTrack3D, a patented 3D gesture-control system for developers, original equipment manufacturers and public-display providers. Their software development kit comes with a set of one-handed and two-handed gestures in the library.

Of course, sometimes you do not have the budget to swap out existing displays for newer ones. In such cases, DISPLAX Skin Multitouch is a patent-pending technology that lets users turn non-conductive flat surfaces into multi-touch surfaces that are liquid and ultraviolet rays resistant.

DISPLAX features a proprietary noise-cancelling technology called XTR-Shield that helps reduce the gap between the sensor and the display, which also helps remove parallax. It uses a projected capacitive technology and can handle up to 20 simultaneous touches with a relative accuracy of one millimetre.

PQ Labs G4 overlay is another component with a similar use case that can be fixed in front of a television or monitor to change it into a multi-touch display.

Fingers gesturing above a Project Soli chip by Google

